Proposal: OFDM Channel Reconstruction

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2. Title for the project

OFDM Channel Reconstruction from Sparse Noisy Pilots Using Measured CSI

3. Description of the dataset

- **Dataset Name:** CSI Dataset dichasus-005x: Indoor Non-Line of Sight, Single Room, a public set of real wireless channel measurements collected by the University of Stuttgart with the DICHASUS channel sounder.
- **Data Source and Access:** Florian Euchner and Marc Gauger published the dataset on the DaRUS research data repository in 2021 under the CC BY 4.0 license (DOI: 10.18419/darus-2204, available at https://doi.org/10.18419/darus-2204). The repository provides five TFRecord files and a machine-readable specification of the measurement parameters.
- **Measurement Environment:** A transmitter mounted on a vacuum robot moved through an indoor room in which a metallic barrier blocks the line of sight, so the data capture reflected paths of a real indoor environment. The system uses a 1.272 GHz carrier frequency, 50 MHz bandwidth and 32 co-located receive antennas; transmitter positions come from the robot's LiDAR.
- **Sample Size and Structure:** There are 26,161 snapshots from five complete robot runs, about 6.9 GB in total. Each snapshot contains a CSI tensor of shape (32, 1024, 2): 32 antennas, 1,024 OFDM subcarriers stored in FFT order, and the real and imaginary parts of each coefficient.
- **Key Features:** The csi field holds the frequency-domain channel coefficients, snr an SNR estimate for each antenna, pos-lidar the two-dimensional transmitter position and time the timestamp. Estimated per-antenna phase and timing offsets of the sounder are published with the dataset and will be compensated.
- **Inputs and Reconstruction Target:** We use one block of 64 subcarriers (3.1 MHz) inside the occupied band, away from the DC subcarrier and the band edges. Pilots occupy 16, 8 or 4 approximately uniformly spaced subcarriers, including both block ends. To emulate least-squares pilot estimates, complex Gaussian noise is added to the measured CSI at the pilot subcarriers, with a pilot SNR of 0–30 dB defined relative to the average channel power of the training data. The remaining subcarriers are the reconstruction targets, and the measured CSI serves as the reference.
- **Dataset Partition and Quality:** Runs 0053–0055 (15,940 snapshots) are used for training, run 0056 (5,220) for validation and run 0057 (5,001) for testing, so adjacent records of one run never fall into different sets. Each of the 32 antennas of a snapshot gives a separate example. Vectors whose CSI error, estimated from fourth-order differences across frequency, exceeds −30 dB relative to the channel power are removed before fitting (about 0.5% in our check; the snr field alone did not identify them). All preprocessing parameters and covariance estimates are fitted on training data only.

4. What Problems It Solves

- **Core Problem:** An OFDM receiver estimates the channel at a few pilot subcarriers and must infer it on the others; accuracy is limited by pilot noise and pilot spacing. We study how accurately a block of measured channel coefficients can be reconstructed from a few noisy pilots, i.e. the interpolation and denoising stage of pilot-based channel estimation.

- **Practical Significance:** Pilots occupy resources that could carry data, so knowing the accuracy each pilot density achieves under realistic noise informs the pilot overhead. Classical receivers interpolate least-squares pilot estimates [1] or apply LMMSE with channel statistics [2]; neural estimators such as ChannelNet [3] and ReEsNet [4] were mostly evaluated on simulated channels, so measured channels [5] give a useful test. Our conclusions concern one measured environment, not a demonstrated pilot reduction in hardware.
- **Preliminary Check:** The measured channel is very smooth in frequency (correlation about 0.97 across the whole 64-subcarrier block): with noise-free observations, linear interpolation from 8 of 64 subcarriers already reached about −43 dB NMSE, close to the measurement error, so all methods looked alike. With noisy pilots they separate: on the full training runs and validation run 0056 (test run 0057 untouched), LMMSE beat linear interpolation by about 4–10 dB with 8 or 16 pilots over 0–30 dB pilot SNR (paired 95% CI about ±0.1 dB). We therefore study noisy pilots.
- **Research Question and Objectives:** How do linear and cubic-spline interpolation, covariance-based LMMSE and a small neural network compare as the pilot SNR (0–30 dB) and pilot density (16, 8 or 4 of 64) vary? Because LMMSE with accurate statistics is the best linear estimator, we ask whether and when a nonlinear learned model adds value, for example at low SNR or when the noise level is not known exactly, and at what computational cost.

5. Methods Will be Used

- **Data Preprocessing:** We read the TFRecords with a tested NumPy-only parser, apply fftshift so that subcarriers are in frequency order, compensate the antenna offsets and extract the block, chosen on training data and kept fixed. All channels are scaled by one constant from the training data; normalizing each snapshot by its own true power would reveal the channel norm to the estimators and, in our check, gave the network an apparent 1–2 dB advantage.
- **Conventional Baselines:** Linear and natural cubic-spline interpolation need no training. LMMSE uses the block mean and covariance estimated from the training runs, the known noise variance and a small diagonal loading chosen on validation data; with sample statistics it equals ridge regression from the pilots to the withheld coefficients, linking it to the regularized regression covered in the course.
- **Learning Model:** A multilayer perceptron with two hidden layers of 128 ReLU units maps the real and imaginary parts of the noisy pilots and the pilot SNR to the withheld coefficients (about 33,000 parameters). One model per pilot density is trained over pilot SNRs drawn uniformly from 0–30 dB with fresh noise each epoch, using mean-squared error, Adam and early stopping on run 0056. We also test a loss weighted by the expected error at each SNR and a residual variant that only corrects the LMMSE estimate.
- **Experimental Evaluation:** The primary measure is NMSE on withheld subcarriers (total squared error over total squared reference magnitude), plotted against pilot SNR for each pilot density on test run 0057, with identical pilots, noise realizations and test records for all methods. Neural training uses three seeds, and 95% confidence intervals come from bootstrapping blocks of consecutive test snapshots. We also report parameter counts, CPU inference time and robustness when the assumed noise variance is wrong. If time permits, NMSE will be translated into bit error rate for QPSK and 16-QAM with zero-forcing equalization.
- **Expected Outcomes:** We expect reproducible code and NMSE-versus-SNR curves for all methods. A full-data validation check already shows LMMSE is a strong baseline: the MLP matched it within 0.2 dB up to 20 dB pilot SNR but was 1.0–1.5 dB worse at 30 dB; the weighted loss halved this gap, a 3×256 network did not help, and the residual variant matched LMMSE. Because one block is nearly two-dimensional (two eigenvalues hold 99.99% of the energy), we will, time permitting, also test joint antenna–frequency reconstruction, where nonlinear structure may matter. Limitations: one indoor environment, synthetic pilot noise, and the reference CSI's own error (median about −43 dB).

References: [1] S. Coleri et al., IEEE Trans. Broadcast. 48(3):223–229, 2002. [2] O. Edfors et al., IEEE Trans. Commun. 46(7):931–939, 1998. [3] M. Soltani et al. (ChannelNet), IEEE Commun. Lett. 23(4):652–655, 2019. [4] L. Li et al. (ReEsNet), IEEE Wireless Commun. Lett. 9(5):615–618, 2020. [5] F. Euchner et al. (DICHASUS), WSA 2021.